

eSurvey Report:

Docker and Containerization in the UNT CS Curriculum

Dr. Cynthia Pope, TECM 4180 | University of North Texas

OPENING STATEMENT

This report examines student awareness of and experience with Docker and containerization tools within the University of North Texas (UNT) Computer Science curriculum. Data was gathered through a Qualtrics eSurvey distributed to UNT CS students during the spring 2026 semester. After analyzing 28 responses, the data revealed three consistent findings: the majority of CS students lack formal exposure to Docker through their coursework, most students feel underprepared for the DevOps expectations they encounter in internship and job searches, and an overwhelming proportion of respondents believe containerization should be formally integrated into the UNT CS curriculum. These patterns indicate a meaningful gap between what the curriculum currently offers and what industry demands, and they provide a strong empirical foundation for the proposed curricular update outlined in Unit 3.

BACKGROUND AND METHODOLOGY

This eSurvey employed primary research methods to collect direct input from the target audience: current UNT Computer Science students, with a particular emphasis on juniors and seniors approaching internship and job applications. The survey was distributed across multiple channels, including Canvas email to classmates in TECM 4180 and other UNT CS courses, group chats with peers from current and prior CS courses, the shared class Google Document on Canvas, and LinkedIn outreach to UNT CS students and recent graduates. Additionally, a QR code linked to the survey was posted on bulletin boards in the CS and engineering buildings on the UNT Denton campus.

A total of 28 responses were collected over a one-week period. While this sample represents a meaningful snapshot of student opinion, it should be noted that 28 respondents constitute a small fraction of the total UNT CS student population; therefore, conclusions drawn from this data are categorical and directional rather than statistically generalizable. Respondents skewed toward upperclassmen, which aligns with the survey's intended audience but limits representation of freshman and sophomore perspectives. No personally identifiable information was collected, and participation was entirely voluntary.

FINDINGS

Finding 1: Most Students Have Not Used Docker Through UNT Coursework

When asked whether they had ever used Docker or a containerization tool, 17 of 28 respondents (61%) answered yes — however, the follow-up question revealed a critical detail: of those 17, **none** identified a UNT course as their primary learning source. Instead, respondents credited personal or side projects (8), online self-study (6), and internships or jobs (3). This finding suggests that while many CS students do encounter Docker, they are largely acquiring this skill independently rather than through the curriculum. The remaining 11 respondents (39%) reported no experience with containerization tools at all. See Figure 1.

**Figure 1: Prior Docker / Containerization Experience
(n = 28)**

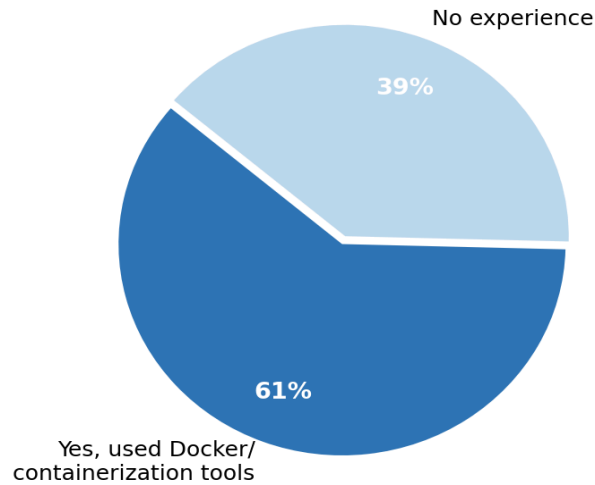
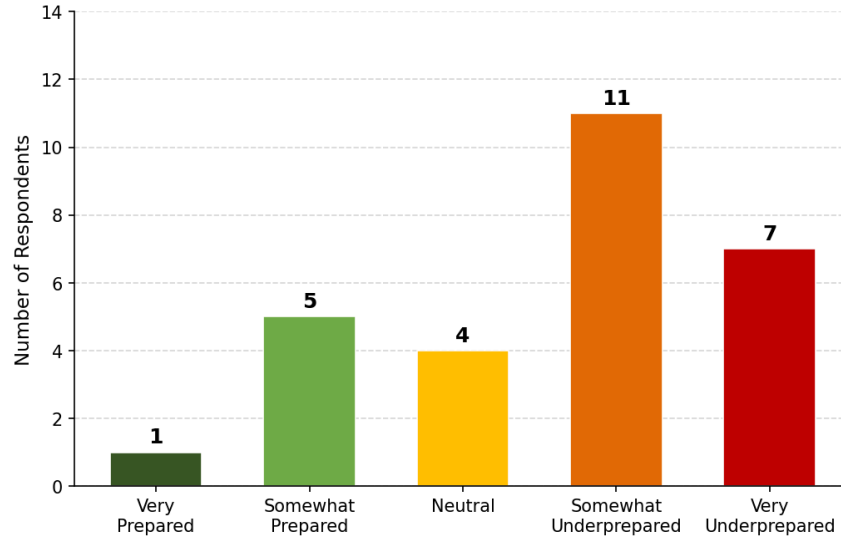


Figure 1: Prior Docker/Containerization Experience (n = 28)

When asked specifically whether any UNT CS course had covered Docker as part of its curriculum, 18 respondents stated it had never been covered, and 7 said it was only briefly mentioned without any depth of instruction. Only 3 respondents indicated they had encountered it in even one course. These findings collectively confirm that containerization remains largely absent from formal UNT CS instruction.

Finding 2: Students Feel Underprepared for DevOps Expectations

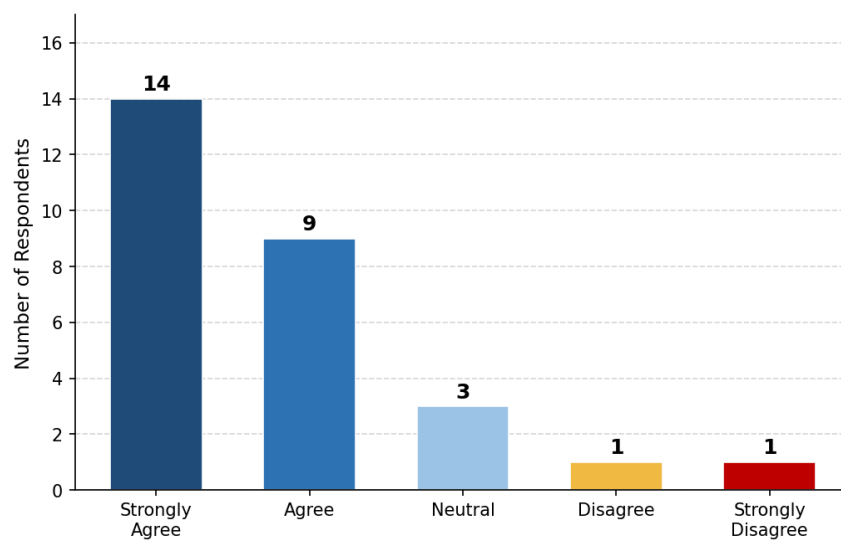
Respondents expressed significant dissatisfaction with how well the UNT CS curriculum prepares them for real-world DevOps tools and practices. As shown in Figure 2, 18 of 28 respondents (64%) indicated they feel either somewhat underprepared or very underprepared for the DevOps expectations they encounter in internship and job applications. Only 6 respondents felt at least somewhat prepared, and a single respondent described themselves as very prepared. This distribution reflects a curriculum that may adequately develop core theoretical competencies but leaves students without the practical, industry-relevant tooling experience employers increasingly expect on day one.

**Figure 2: Student Preparedness for DevOps Tools
(n = 28)***Figure 2: Student Preparedness for DevOps Tools and Practices (n = 28)*

Supporting this finding, 20 of 28 respondents (71%) reported having encountered a job posting or internship listing that required Docker or containerization knowledge, with 9 saying they see such requirements frequently. This confirms that the skills gap is not hypothetical — students are directly experiencing it as they apply for opportunities.

Finding 3: Strong Student Support for Formal Docker Instruction

Despite the current curriculum gap, student enthusiasm for a formal solution is clear. As illustrated in Figure 3, 23 of 28 respondents (82%) agreed or strongly agreed that Docker and containerization should be formally taught as part of the UNT CS curriculum. Only 2 respondents expressed any disagreement. This level of consensus is notable and suggests that a curricular update would be received positively by the student body.

**Figure 3: Should Docker Be Formally Taught at UNT?
(n = 28)***Figure 3: Should Docker Be Formally Taught at UNT? (n = 28)*

The open-ended question asking students which tool they most wished the program had introduced yielded responses that further reinforced this finding. Docker was the most commonly mentioned tool by name, appearing in 9 of the 28 open-ended responses. Other frequently cited tools included Git/GitHub workflows (5), CI/CD pipelines (4), and Linux command-line fundamentals (3). Several respondents used terms like *"industry-standard"* and *"expected me to already know it"* when describing Docker — a sentiment that underscores the urgency of the proposed improvement.

CONCLUSION

This eSurvey report examined how UNT Computer Science students perceive the current curriculum's coverage of Docker and containerization tools and whether they believe formal instruction in these areas is warranted. Three findings emerged consistently from the 28 responses analyzed: most students have no formal coursework exposure to Docker, the majority feel underprepared for the DevOps skills employers require, and an overwhelming proportion support integrating containerization into the curriculum. These findings align directly with the argument advanced in the Unit 3 Proposal and provide empirical support for the case that a curricular update is both needed and desired. Given the strength and consistency of the data — particularly the 82% support rate for formal Docker instruction — this eSurvey suggests the proposed improvement has a strong likelihood of being well-received by students, faculty, and departmental stakeholders alike. The data will be used to inform the charts, analysis, and rhetorical framing of the Unit 4 Formal Report.

WORKS CITED

Saspara, Meetkumar. "Docker and Containerization in the UNT CS Curriculum." Qualtrics eSurvey. Distributed via Canvas, LinkedIn, group chats, and QR code, Feb.–Mar. 2026.